National Testing Agency

Question Paper Name :	B Tech 6th Apr 2026 Shift 1
Subject Name :	B. Tech
Creation Date :	2026-04-06 14:36:27
Duration :	180
Total Marks :	300
Display Marks:	Yes

B. Tech

Group Number :	1
Group Id :	69527828
Group Maximum Duration :	0
Group Minimum Duration :	180
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	300

Mathematics Section A

Section Id :	695278151
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	695278151
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 6952782136 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x)=\sin^{-1}\left(\frac{x+[x]}{3}\right)$ is $[\alpha, \beta)$, then $\alpha^2+\beta^2$ is equal to:

Options :

6952787496. 2

6952787497. 5

6952787498. 10

6952787499. 13

Question Number : 2 Question Id : 6952782137 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let one root of the quadratic equation in x:
$(k^2 - 15k + 27)x^2 + 9(k - 1)x + 18 = 0$
be twice the other. Then the length of the latus rectum of the parabola $y^2 = 6kx$ is equal to:

Options :

6952787500. 4

6952787501. 6

6952787502. 8

6952787503. 12

Question Number : 3 Question Id : 6952782138 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let e_1 and e_2 be two distinct roots of the equation $x^2 - ax + 2 = 0$. Let the sets
$\{a \in \mathbb{R} : e_1$ and e_2 are the eccentricities of hyperbolas$\} = (\alpha, \beta)$, and
$\{a \in \mathbb{R} : e_1$ and e_2 are the eccentricities of an ellipse and a hyperbola, respectively$\} = (\gamma, \infty)$.
Then $\alpha^2 + \beta^2 + \gamma^2$ is equal to:

Options :

6952787504. 18

6952787505. 22

6952787506. 26

6952787507. 34

Question Number : 4 Question Id : 6952782139 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the set of all values of $k \in \mathbb{R}$ such that the equation $z(\bar{z} + 2 + i) + k(2 + 3i) = 0$, $z \in \mathbb{C}$, has at least one solution, be the interval $[\alpha, \beta]$. Then $9(\alpha + \beta)$ is equal to:

Options :

6952787508. -10

6952787509. -8

6952787510. $10\sqrt{13}$

6952787511. $8\sqrt{13}$

Question Number : 5 Question Id : 6952782140 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The value of $1^3 - 2^3 + 3^3 - \ldots + 15^3$ is:

Options :

6952787512. 1706

6952787513. 1856

6952787514. 1982

6952787515. 2403

Question Number : 6 Question Id : 6952782141 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The sum of the first ten terms of an A.P. is 160 and the sum of the first two terms of a G.P. is 8. If the first term of the A.P. is equal to the common ratio of the G.P. and the first term of the G.P. is equal to common difference of the A.P., then the sum of all possible values of the first term of the G.P. is:

Options :

6952787516. $\frac{34}{9}$

6952787517. $\frac{34}{13}$

6952787518. $\frac{32}{9}$

6952787519. $\frac{32}{13}$

Question Number : 7 Question Id : 6952782142 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of 4-letter words, with or without meaning, each consisting of two vowels and two consonants that can be formed from the letters of the word INCONSEQUENTIAL, without repeating any letter, is:

Options :

6952787520. 2670

6952787521. 2840

6952787522. 2920

6952787523. 3600

Question Number : 8 Question Id : 6952782143 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the coefficients of the middle terms in the binomial expansions of $(1 + \alpha x)^{26}$ and $(1 - \alpha x)^{28}$, $\alpha \neq 0$, are equal, then the value of α is:

Options :

6952787524. 1

6952787525. $\frac{14}{13}$

6952787526. $\frac{27}{7}$

6952787527. $\frac{7}{27}$

Question Number : 9 Question Id : 6952782144 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A data consists of 20 observations $x_1, x_2, ..., x_{20}$. If $\sum_{i=1}^{20}(x_i+5)^2 = 2500$ and $\sum_{i=1}^{20}(x_i-5)^2 = 100$, then the ratio of mean to standard deviation of this data is:

Options :

6952787528. 2:1

6952787529. 3:1

6952787530. 3:2

6952787531. 4:1

Question Number : 10 Question Id : 6952782145 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A bag contains (N + 1) coins − N fair coins, and one coin with 'Head' on both sides. A coin is selected at random and tossed. If the probability of getting 'Head' is $\frac{9}{16}$, then N is equal to:

Options :

6952787532. 5

6952787533. 7

6952787534. 8

6952787535. 9

Question Number : 11 Question Id : 6952782146 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the eccentricity e of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, passing through $\left(6, 4\sqrt{3}\right)$, satisfies $15(e^2 + 1) = 34e$, then the length of the latus rectum of the hyperbola $\frac{x^2}{b^2}-\frac{y^2}{2\left(a^2+1\right)}=1$ is:

Options :

6952787536. 10

6952787537. 20

6952787538. 25

6952787539. 30

Question Number : 12 Question Id : 6952782147 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let chord PQ of length $3\sqrt{13}$ of the parabola $y^2 = 12x$ be such that the ordinates of points P and Q are in the ratio 1:2. If the chord PQ subtends an angle α at the focus of the parabola, then $\sin\alpha$ is equal to:

Options :

6952787540. $\frac{3}{5}$

6952787541. $\frac{4}{5}$

6952787542. $\frac{5}{13}$

6952787543. $\frac{12}{13}$

Question Number : 13 Question Id : 6952782148 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $0 < \alpha < 1, \beta = \frac{1}{3\alpha}$ and $\tan^{-1}(1-\alpha) + \tan^{-1}(1-\beta) = \frac{\pi}{4}$. Then $6(\alpha + \beta)$ is equal to:

Options :

6952787544. 6

6952787545. 7

6952787546. 8

6952787547. 9

Question Number : 14 Question Id : 6952782149 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $S = \left\{\theta \in (-2\pi, 2\pi) : \cos\theta + 1 = \sqrt{3}\sin\theta\right\}$.

Then $\sum_{\theta \in S} \theta$ is equal to:

Options :

6952787548. $-\frac{2\pi}{3}$

6952787549. $-\frac{4\pi}{3}$

6952787550. $\frac{2\pi}{3}$

6952787551. $\frac{4\pi}{3}$

Question Number : 15 Question Id : 6952782150 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the image of the point P(1, 6, a) in the line $L: \frac{x}{1} = \frac{y-1}{2} = \frac{z-a+1}{b}, b > 0,$ be $\left(\frac{a}{3}, 0, a+c\right)$. If S(α, β, γ), α > 0, is the point on L such that the distance of S from the foot of perpendicular from the point P on L is $2\sqrt{14}$, then α + β + γ is equal to:

Options :

6952787552. 19

6952787553. 20

6952787554. 21

6952787555. 22

Question Number : 16 Question Id : 6952782151 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let a line L be perpendicular to both the lines $L_1: \frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $L_2: \frac{x-2}{1} = \frac{y-4}{4} = \frac{z-6}{7}$.

If θ is the acute angle between the lines L and $L_3: \frac{x-\frac{8}{7}}{2} = \frac{y-\frac{4}{7}}{1} = \frac{z}{2}$, then tan θ is equal to:

Options :

6952787556. $\frac{3}{2}\sqrt{2}$

6952787557. $\frac{5}{2}\sqrt{2}$

6952787558. $\frac{5}{3}\sqrt{2}$

6952787559. $\frac{4}{3}\sqrt{2}$

Question Number : 17 Question Id : 6952782152 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The value of $\lim_{x \to 0} \left(\frac{x^2 \sin^2 x}{x^2 - \sin^2 x}\right)$ is:

Options :

6952787560. 2

6952787561. 3

6952787562. 4

6952787563. 6

Question Number : 18 Question Id : 6952782153 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\frac{32\cos^4 x}{1+e^{\sin x}} \right) dx$ is:

Options :

6952787564. $4\pi + 2$

6952787565. $3\pi + 8$

6952787566. $3\pi + 4$

6952787567. $4\pi + 3$

Question Number : 19 Question Id : 6952782154 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The area of the region $\{(x, y) : 0 \le y \le 6 - x, y^2 \ge 4x - 3, x \ge 0\}$ is:

Options :

6952787568. 8

6952787569. 9

6952787570. 12

6952787571. 15

Question Number : 20 Question Id : 6952782155 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let e be the base of natural logarithm and let $f : \{1, 2, 3, 4\} \to \{1, e, e^2, e^3\}$ and $g : \{1, e, e^2, e^3\} \to \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}\right\}$ be two bijective functions such that f is strictly decreasing and g is strictly increasing. If $\phi(x) = \left[f^{-1}\left\{ g^{-1}\left(\frac{1}{2}\right)\right\}\right]^x$, then the area of the region R = $\{(x, y): x^2 \le y \le \phi(x), 0 \le x \le 1\}$ is:

Options :

6952787572. $\frac{3 - \log_e(2)}{3\log_e(2)}$

6952787573. $\frac{1}{3\log_e(2)}$

6952787574. $3 + \log_e(2)$

6952787575. $\frac{3 + \log_e(2)}{2 + \log_e(3)}$

Mathematics Section B

Section Id :	695278152
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	695278152
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 21 Question Id : 6952782156 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $A=\begin{bmatrix}-1 & 1 & -1\\ 1 & 0 & 1\\ 0 & 0 & 1\end{bmatrix}$ satisfy

$$A^2+\alpha\left(adj\left(adj(A)\right)\right)+\beta\left(adj(A)\left(adj\left(adj(A)\right)\right)\right)=\begin{bmatrix}2 & -2 & 2\\ -2 & 0 & -1\\ 0 & 0 & -1\end{bmatrix}$$ for

some $\alpha, \beta \in \mathbb{R}$.

Then $(\alpha-\beta)^2$ is equal to _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 22 Question Id : 6952782157 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let the centre of the circle $x^2+y^2+2gx+2fy+25=0$ be in the first quadrant and lie on the line $2x-y=4$. Let the area of an equilateral triangle inscribed in the circle be $27\sqrt{3}$. Then the square of the length of the chord of the circle on the line $x=1$ is_____.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 23 Question Id : 6952782158 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{j}-\hat{k}$ and $\vec{c}$ be three vectors such that $\vec{a}\times\vec{c}=\vec{b}$ and $\vec{a}\cdot\vec{c}=3$, then $\vec{c}\cdot\left(\vec{a}-2\vec{b}\right)$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 24 Question Id : 6952782159 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

For the functions $f(\theta) = \alpha \tan^2\theta + \beta \cot^2\theta$, and

$g(\theta) = \alpha \sin^2\theta + \beta \cos^2\theta$, $\alpha > \beta > 0$, let $\min\limits_{0<\theta<\frac{\pi}{2}} f(\theta) = \max\limits_{0<\theta<\pi} g(\theta)$. If the first term of a G.P. is $\left(\frac{\alpha}{2\beta}\right)$, its common ratio is $\left(\frac{2\beta}{\alpha}\right)$ and the sum of its first 10 terms is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 25 Question Id : 6952782160 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $y = y(x)$ be the solution of the differential equation $\left(x^2 - x\sqrt{x^2-1}\right)dy + \left(y\left(x - \sqrt{x^2-1}\right) - x\right)dx = 0$, $x \geq 1$. If $y(1)=1$, then the greatest integer less than $y\left(\sqrt{5}\right)$ is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Physics Section A

Section Id :	695278153
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	695278153
Question Shuffling Allowed :	Yes
Is Section Default? :	No